

Documentation & Deployment:

1. Report Overview:

The Airbnb Analytics Dashboard provides insights into listing performance, pricing trends, host activity, review metrics, property availability, and revenue forecasting through What-If scenario analysis

2. Report Navigation

Page 1 – Executive Dashboard

View high-level KPIs.
Monitor average pricing and availability.
Track projected revenue.

Key Visuals : Total Listing, Total Reviews, Total Hosts, Average Price, Project Revenue, Average Occupancy rate.

Page 2 – Pricing & Geographic Analysis

Analyze pricing distribution across properties.
Explore property locations using maps.

Key Visuals : Property Location Map, Price Histogram, Average Price by Property type.

Page 3 – Scenario Analysis

Evaluate business impact under different occupancy and seasonal conditions.

Key Visuals : Scenario Revenue, Occupancy Rate, Seasonal Multiplier Impact.

Listing Detail Page (Drill-Through)

Investigate individual listing performance.

Information Available:

Listing Details
Host Information
Review Metrics
Availability Statistics
Revenue Projections

Documentation:

1. Data Model Tables:

Hosts
Listings
Date Table
Calendar
Reviews

Relationships
Hosts[host_id] → Listings[host_id]
Listings[listing_id] → Calendar[listing_id]
Listings[listing_id] → Reviews[listing_id]

Relationship Type:
One-to-Many

Single Direction Filter

2. Power Query Transformations

Calendar Table

Transformations:

Remove currency symbols from Price.
Remove commas from numeric fields.
Convert Available field to Boolean.
Convert Date field to Date datatype.
Replace null Minimum Nights values.
Replace null Maximum Nights values.

Listings Table

Transformations:

Extract numeric Bathrooms value.
Standardize Property Type.
Standardize Room Type.
Convert Latitude and Longitude to Decimal.
Create Amenity Lookup Table.

Hosts Table

Transformations:

Convert Host Since to Date.
Create Host Biography indicators.

Reviews Table

Transformations:

Convert Review Date to Date.
Create Review Length metric.

3. DAX Measures :

Pricing Metrics
Pricing Metrics
Review Metrics
Host Metrics
Revenue Metrics

4. What-If Parameter Configuration

Occupancy Rate Parameter
Seasonal Multiplier Parameter